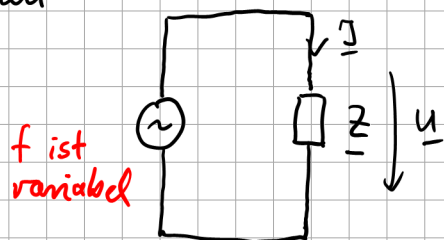


8 Netzwerke mit veränderlicher Frequenz

neu:



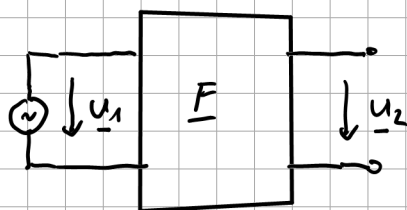
z.B.



$$\underline{Z}_C = \frac{1}{j\omega C}$$

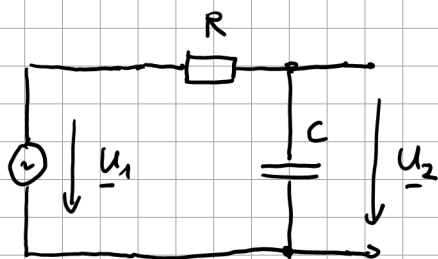
$$\underline{I}_C = \frac{\underline{U}}{\underline{Z}_C} = \underline{U} \cdot j\omega C$$

Systemtheorie:



$$\underline{U}_1 \cdot \underline{F} = \underline{U}_2$$

Bsp:



mit $\underline{F} = \underline{F}(j\omega)$

$$\Rightarrow \underline{F} = \frac{\underline{U}_2}{\underline{U}_1} \quad (\text{Führungsübertragungsfkt.})$$

$$\underline{A}(j\omega) = \frac{\underline{U}_2}{\underline{U}_1} = \frac{\underline{Z}_C}{R + \underline{Z}_C} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \cdot \frac{j\omega C}{j\omega C} = \frac{1}{1 + j\omega RC}$$

Übertragungsfunktion "Amplifier"

$$\underline{A}(j\omega) = \underbrace{|\underline{A}(j\omega)|}_{\text{Betrag}} \cdot \underbrace{e^{j\varphi}}_{\text{Winkelfunktion}}$$

Idee: zur Darstellung: $|A(j\omega)|$, φ_A getrennt darstellen

$$\underline{A}(j\omega) = \frac{1}{1+j\omega RC}$$

Betrag:

$$|A(j\omega)| = \left| \frac{1}{1+j\omega RC} \right|$$

Mathe: $\left| \frac{2+3i}{5-2i} \right| = \frac{|2+3i|}{|5-2i|} = \frac{\sqrt{2^2+3^2}}{\sqrt{5^2+2^2}}$

$$= \frac{1}{|1+j\omega RC|} = \frac{1}{\sqrt{1^2 + (\omega RC)^2}} = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

Amplitudengang $\hat{=} |A| = f(\omega)$

Winkel:

$$\varphi_A = \angle \left(\frac{1}{1+j\omega RC} \right) = \varphi_{\text{Zähler}} - \varphi_{\text{Nenner}}$$

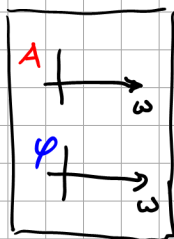
$$= \angle(1) - \angle(1+j\omega RC)$$

$$= 0^\circ - \arctan\left(\frac{\omega RC}{1}\right)$$

$$= -\arctan(\omega RC)$$

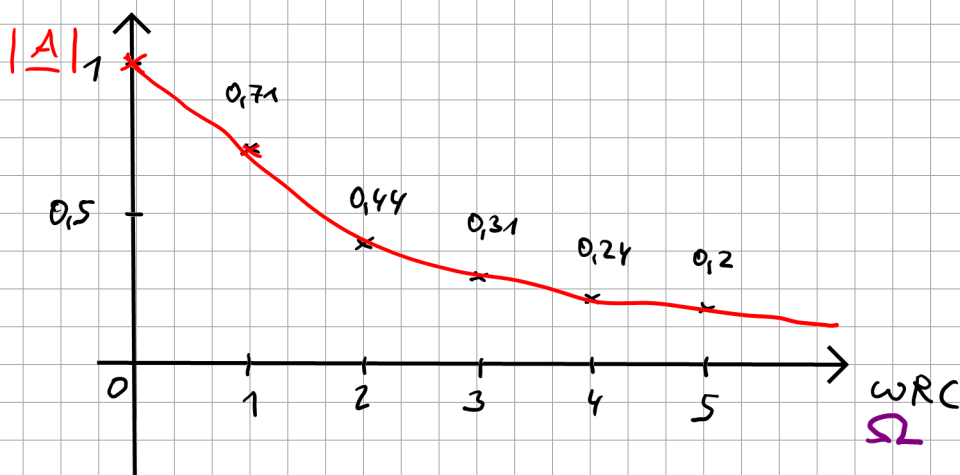
Phasengang $\hat{=} \varphi_A = f(\omega)$

Ziel:

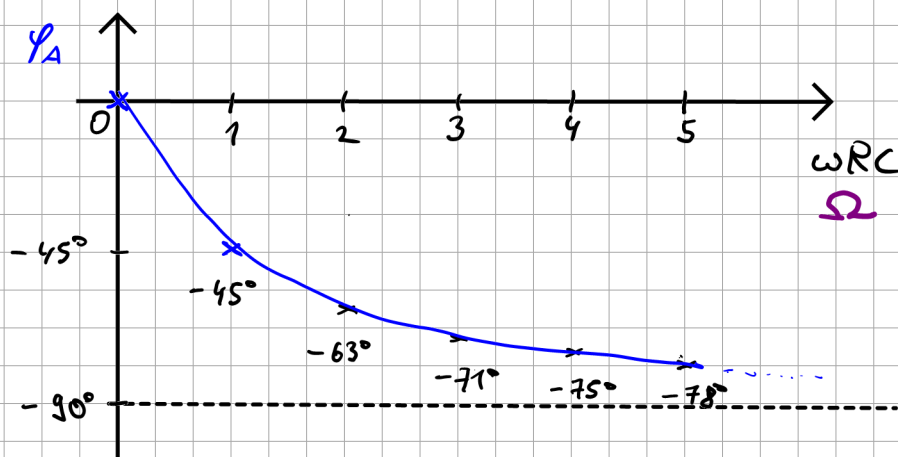


$$|A(j\omega)|$$

$$= \frac{1}{\sqrt{1 + (\underbrace{\omega RC}_{\Omega})^2}}$$



$$\varphi_A = -\arctan(\underbrace{\omega RC}_{\Omega})$$



Neue Darstellungsform für frequenzabhängige Diagramme

=> „Bode - Diagramm“

Herr Bode : Elektroingenieur (Bell Company) ~ 1930

Normmeß : Dezibel

$$\begin{array}{c} \swarrow \quad \searrow \\ \text{„10“} \quad \frac{|u_2|}{|u_1|} \hat{=} \text{„bel“} \end{array}$$

$$a_{dB} = 20 \cdot \log_{10} |A(j\omega)|$$

Amplitudenmeß in dB

$$\begin{aligned} & 20 \log_{10} |A(\Omega)| \\ &= 20 \log_{10} | \underbrace{k}_{\uparrow} \cdot |A(\Omega)| | \end{aligned}$$

$$\begin{aligned} & \log(a \cdot b) \\ &= \log a + \log b \end{aligned}$$

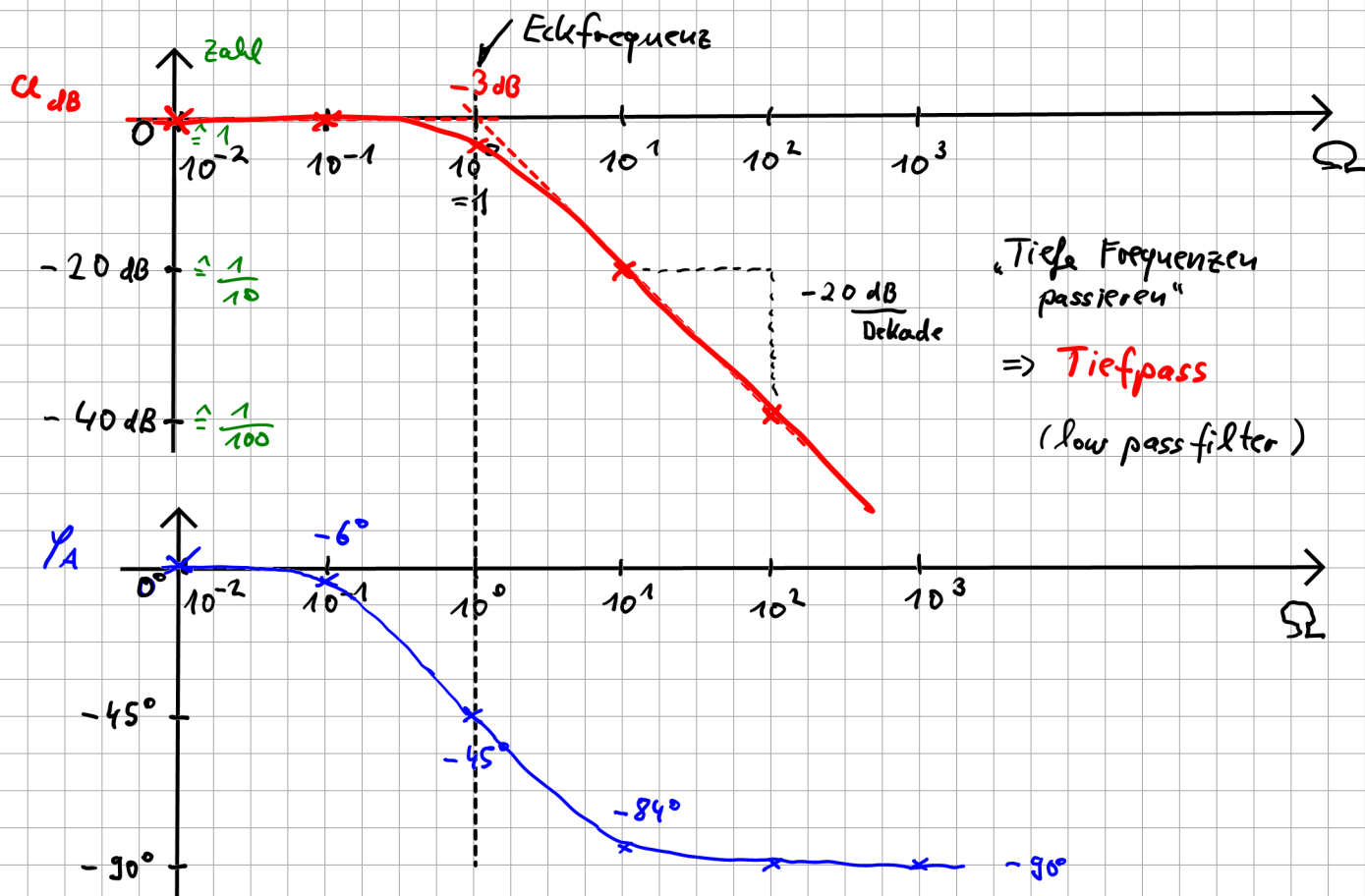
$$|A(j\omega)| = \frac{1}{\sqrt{1 + \Omega^2}}$$

$$\varphi_A = -\arctan \Omega$$

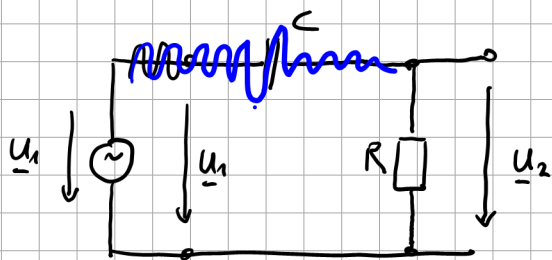
$$a_{dB} = 20 \cdot \log_{10} |A|$$

Ω	$\log_{10}(\Omega)$	$ A(j\omega) $	a_{dB} / dB	φ
$10^{-2} = 0,01$	-2	1	0	-0,6°
$10^{-1} = 0,1$	-1	0,995	-0,04	-5,7°
$\frac{1}{4}$	-0,6	0,97	-0,26	-14°
$\frac{1}{2}$	-0,3	0,89	-0,97	-26,6°
1	0	0,707	-3	-45°
2	0,3	0,45	-6,99	-63,4°
4	0,6	0,243	-12,3	-76°
10	1	0,1	-20	-84°
100	2	0,01	-40	-90°

is hier wichtig



RC - Hochpass



$$\underline{A}(j\omega) = \frac{U_2}{U_1} = \frac{R}{R + \underline{Z}_C} = \frac{R}{R + \frac{1}{j\omega C}} = \frac{j\omega C R}{1 + j\omega C R}$$

$$\underline{A}(j\omega) = \frac{j\Omega}{1 + j\Omega}$$

Amplitudengang:

$$|A| = \left| \frac{j\omega RC}{1 + j\omega RC} \right| = \frac{|j\omega RC|}{|1 + j\omega RC|} = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}}$$

Phasengang

$$\begin{aligned} \angle \left(\frac{j\omega RC}{1 + j\omega RC} \right) &= \varphi_{\text{Zähler}} - \varphi_{\text{Nenner}} = \angle(j\omega RC) - \angle(1 + j\omega RC) \\ &= 90^\circ - \arctan\left(\frac{\omega RC}{1}\right) \\ &= 90^\circ - \arctan(\omega RC) \end{aligned}$$

$$\begin{aligned} |A(j\omega)| \\ = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}} \end{aligned}$$

Ω	$ A(j\omega) $	a_{dB}	φ_A
10^{-2}	0,01		
10^{-1}	0,1		
$\frac{1}{2}$	0,45		
1	$\frac{1}{\sqrt{2}}$		
2	0,89		
10^1	0,995		
10^2	1		